

Hsuan-Yu (Marco) Wang

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Professional Summary

Machine learning engineer and data scientist with experience in computational linguistics, speech processing, and NLP systems, with hands-on experience in end-to-end ML pipelines, conversational AI applications, and multimodal speech systems.

Technical Skills

Programming Languages: Python, C/C++, JavaScript, SQL, R, Swift
Machine Learning & Deep Learning: PyTorch, Transformers, Scikit-learn, Retrieval-Augmented Generation, Diffusion Models, LangChain, LangGraph, ReAct, CLIP, Large Language Models, Vision-Language Models, Multimodal Learning
Speech & Audio Processing: ESPnet2, SpeechBrain, WhisperX, pyannote.audio, Dialogflow, Wav2Vec 2.0, Automatic Speech Recognition, Text-to-Speech, Speech Emotion Recognition, Speaker Diarization
Data Science & Analysis: NumPy, Pandas, Matplotlib, PostgreSQL, Seaborn
Experimentation, Backend & Deployment: FastAPI, Docker, AWS, Google Cloud Platform, Git, Linux, Conda, Weights & Biases

Work Experience

Graduate Research Assistant Aug 2025 – Present
Speech, Language and Interactive Media Lab, University of British Columbia

- Research diffusion-based approaches for text-to-speech systems by building end-to-end experimentation workflows covering dataset curation, preprocessing, model training, inference, and evaluation.
- Implemented block diffusion mechanisms into diffusion language model-style TTS framework to accelerate inference by processing tokens in parallel blocks, and extended TTS model into a dLLM-based TTS system by integrating Edit-Flow methods within the dLLM library to enable token-level editing and controllable speech generation.

Projects

Healthcare Conversational AI Summarization System Mar 2026

- Developing a healthcare conversational AI system that records, transcribes, diarizes, and summarizes multi-speaker clinical consultations while reducing hallucinations in generated summaries.
- Integrating WhisperX, Retrieval-Augmented Generation, and LLM-based summarization pipelines using FastAPI and Google Cloud Platform to support accurate action-item extraction and privacy-conscious local medical data processing workflows.

Annotated Corpus Platform with Interactive Interface Mar 2026

- Collaborated in a group of four to build an end-to-end annotated corpus system with a FastAPI-based interactive interface, supporting data annotation workflows and backend services for scalable data management.
- Containerized the system with Docker for reproducibility and deployment, and extended the platform to support model training and evaluation pipelines for under-resourced languages using the annotated corpus.

Multimodal AI Commerce Agentic System Mar 2026

- Developed a multimodal AI commerce assistant using LangChain, OpenAI APIs, and Retrieval-Augmented Generation (RAG) to support intelligent product search, recommendation, and conversational interactions.
- Integrated CLIP-based vision-language retrieval, vector database search, and tool-calling workflows to enable multimodal understanding and context-aware AI agent behavior across e-commerce scenarios.

Automated Pipeline for TTS Quality Assessment via Speaker Verification Analysis Dec 2025

- Designed an automated evaluation pipeline for F5-TTS on 2,000+ LibriTTS audio segments, integrating SpeechBrain ECAPA-TDNN to extract speaker embeddings and quantify voice cloning quality using cosine similarity.
- Built a scalable workflow for preprocessing, batch inference, and metric reporting, enabling efficient and reproducible evaluation of speaker preservation in zero-shot voice cloning.

Speech Emotion Recognition with ASR-Diarization Alignment May 2025

- Developed a speech emotion recognition (SER) pipeline integrating WhisperX and pyannote.audio outputs into an alignment framework that combines aligned audio segments with Wav2Vec 2.0 and RoBERTa through bimodal cross-attention fusion.
- Achieved a 10% accuracy improvement on the IEMOCAP dataset, demonstrating the effectiveness of aligned and speaker-cleaned segments for SER in multi-speaker conversational settings.

Education

University of British Columbia Vancouver, BC
Master of Data Science in Computational Linguistics Sep 2025 - Jun 2026

National Taiwan Normal University Taipei, TW
Bachelor of Science in Computer Science and Bachelor of Arts in English Sep 2020 - Jun 2025

Publication

“Enhancing Speech Emotion Recognition Leveraging Aligning Timestamps of ASR Transcripts and Speaker Diarization.”
International Conference on Asian Language Processing 2025, pp. 85–90. Hsuan-Yu Wang, Pei-Ying Lee, Berlin Chen.
10.1109/IALP68296.2024.11156415